

# Replication Report: (title not recorded)

OSTI 2571540 | Coverage 8/10, Agreement 8/10

Rick Stevens

Ollie (AI Assistant)  
Argonne National Laboratory

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## 1 Source paper

- **Title:** (title not recorded)
- **Authors:** Not recorded
- **Year:** n/a
- **Domain:** Not recorded
- **Identifier:** OSTI 2571540

## 2 Replication objective

The central claim of the paper concerns not recorded. Our objective was to reproduce the head-line numerical/qualitative results within the constraints of the AI-driven Replication Project (24–72h, commodity GPU/CPU compute, no author contact). A replication plan is archived at `2571540-A-Portfolio-Approach-to-Massively-Parallel/replication_plan.pdf`.

## 3 Methodology

We followed the standard Replication Project workflow: ingest the source PDF, extract method and data pointers, draft a replication plan, attempt the smallest end-to-end pipeline that produces a comparable number, then scale up where compute and data permit. Tools used in this replication are catalogued in the meta-paper’s computational tools inventory (see `COMPUTATIONAL_TOOLS_INVENTORY.md`).

This paper went through the Tier-Lift pipeline; supplementary notes at `.openclaw/workspace/24h-progress/tier-lift-v2/2571540.md`.

### Paper-directory README excerpt

#### A Portfolio Approach to Massively Parallel Bayesian Optimization

- **\*\*OSTI ID:\*\*** 2571540
- **\*\*Rank:\*\*** #2
- **\*\*Replication Score:\*\*** 10/10

#### # Why This Paper

This work is highly suitable for AI replication as it presents a fully computational, algorithmic methodology for parallel Bayesian optimization, uses benchmark functions and simulated data, and produces clear quantitative performance metrics.

### # Replication Plan

AI would implement the batch Bayesian optimization algorithms, run experiments on benchmark functions, and quantitatively compare performance metrics such as speed and solution quality, replicating the portfolio allocation and hypervolume Sharpe ratio analysis.

### # Status

- [ ] Paper reviewed
- [ ] Data identified
- [ ] Code implemented
- [ ] Results validated

The replication was driven by an OpenClaw agent loop (Claude Opus / GPT-5 class models) issuing shell, Python, and (where applicable) GPU jobs. Compute usage and wall-time for individual papers are summarised in the meta-paper’s resource table; not separately recorded for this paper unless noted in the Tier-Lift artefact. Sub-agents were spawned for replication-plan generation and (for papers in batches 1–4) for tier-lift extension runs.

## 4 What was reproduced

- Not recorded.

## 5 What was missing or incomplete

- Not recorded.

The dominant blocker categories for this paper, mapped onto the meta-paper’s 9-category friction taxonomy (§4), are listed in §?? below.

## 6 Quantitative agreement

Per-quantity numerical comparisons are not separately tabulated in the structured evaluation record for this paper beyond the agreement rationale below; where the rationale references specific numbers, they are reproduced verbatim. A full numerical comparison table is recorded in the per-replication artefacts (see §??). For papers below 8/8, the agreement rationale identifies the specific quantities that diverge.

**Agreement rationale.** Not recorded.

## 7 Score and friction-point tagging

- **Coverage:** 8/10 — Not recorded.
- **Agreement:** 8/10 — Not recorded.

**Friction points encountered (taxonomy tags):**

- None recorded.

## 8 Follow-on questions

- Not recorded.

## 9 Reproducibility pointers

- **Paper directory:** 2571540-A-Portfolio-Approach-to-Massively-Parallel
- **Replication artefacts:**
  - Replication artifacts in paper directory.
- **Replication-dir hint from JSONL:** /Users/stevens/Dropbox/REPLICATE-PROJECT/2571540-A-Portfolio-Approach-to-Massively-Parallel
- **Status:** tier\_lift\_v25\_2026\_04\_27

To re-run, change directory into the paper directory above and consult its README.md for the entry-point script. Most replications follow the convention `replication/run_*.sh` or `replication/<subdir>/run.py`.